Nguyễn Thiện Nhân - 2452879 - HW4

Date:

1. ρ = probability of getting heads.

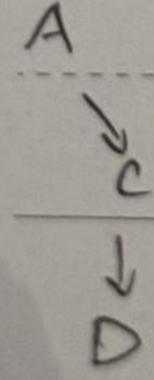
$\Rightarrow L(\rho) \propto \rho^3 (1-\rho)^3$, with coefficient $\binom{12}{9}$ is constant respect to ρ and can be skipped. The likelihood is largest near $\rho = 0,75$, so values of ρ around 0,75 are more plausible than values far from 0,75, but other values are still possible.

2.

Training set contains 4 ~~samples~~ examples with class $Y=0$ and 4 with $Y=1 \rightarrow P(Y=0) = P(Y=1) = 0.5$.

Using add-one smoothing for binary attributes, $P(A=1 \mid Y=1) = \frac{3+1}{4+2} = \frac{4}{6} = \frac{2}{3}$, because 3 class-1 examples have $A=1$, and $P(B=0 \mid Y=1) = \frac{2+1}{4+2} = \frac{1}{2}$. For class 0, $P(A=1 \mid Y=0) = \frac{1+1}{4+2} = \frac{1}{3}$, $P(B=0 \mid Y=0) = \frac{2+1}{4+2} = \frac{1}{2}$. Assuming conditional independence, the unnormalized score for class 1 is $\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{6}$, for class 0 is $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{12}$. Giving $P(Y=1 \mid A=1, B=0) = \frac{1/6}{1/6+1/12} = 2/3$ and $P(Y=0 \mid A=1, B=0) = 1/3 \rightarrow$ Predicted class = 1. It's unrealistic that A and B is independent given Y because it might still be related in real life.

3.



a/ A and B w/o evidence : independent

b/ A and B given C : active / dependent

TEKO SOLUTION

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c/ A and D given C : blocked / d-separated

d/ C and E given B : blocked.

4.

Predictive reasoning moves from ~~the~~ a cause to an expected effect, for example ~~is~~ seeing rain increases the chance that the road is wet.

Diagnostic reasoning moves from an observed effect to a possible cause, for example seeing the wet road makes us think it's more likely rained. Observing a common effect can make two previously independent causes statistically dependent. For example the rain and person watering also makes the garden wet. So if we see a wet garden, and know that it rained, it's less likely that sb watered the gre garden. (explaining away)

5. I'd use Naive Bayes to predict the target class from the features. The model assumes that features are conditionally independent given the class. I would evaluate performance on separate val set and use Brier score to assess probability calibration. A possible failure occurs when correlated features cause model to overestimate probability. Naive Bayes is a simple one effective baseline, but its assumptions should be checked.